

A Packing of 5 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 5 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.461368. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.322896, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 5 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.461368 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 5$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.322896, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 5/4 = 3.461538$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N5.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
576894724940857076060582729967221583637275608833367642905694917073061215
126727071499525189130892630863953305179449565489700032521440989314732871
496443860299677090070650029721118753671136552657
```

denominator:

```
166666666666618246430585198268418831381300600449210232262696559276452023
510226202786965187560959698203581491892529424760542201006699617505201374
722400754173173578226187204629984299014051430952
```

B Coordinates

5 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N5.txt`.

```
255145389961 108486778741 7304003861 344341552484 -149169289219 -243612090212 26797274895 94751205339-107857604147 -75160288363 -509983691765 142359961905
41383118510 -183120886353 310824001449 304578912034 127408999065 288705577566 -43195794713 250139689014 02189391363 -105947652041 -41611356820 -39555322806
-168596471827 -372659510742 -289271671596 -4434336122806 1218433 468298543184 -268346901546 -90807782959004936121 122501264457 462193670715 -163419890022
-619716831838 146368335068 -123287421331 267289005046 7193822905 -204404433541 -92661271266 -794162266561 367582558 -266881649594 -91511988737 9418645251
-64026290398 198367310046 -24868264699 22539331342 64052846772 214175957865 -466606695791 14202685091 25328203477 -598792861078 51461135058 46849702676
-284790064874 267076729877 690090289797 -163378134722 26181605369 133750578006 7781792740 -194933396352 655845656423 -102933922379 286294363368 94769547632
-26330779163 -233295164109 -104152457338 -130323027049 486533258 176232210288 -102754128919 257269811683 44441979866 -94579820468 -309451638936 359972764579
295173262141 120961452633 168095012824 -51839430458 226638230717 -37647876489 -361290534247 1258670021 425667887762 -222183755904 85203170052 495352828856
235538427443 495918286110 -415507681110 -71102851940 1923039614 -343958482199 171776815329 300344787955 52546630287 380592698521 -221558786005 24098011495
-10734788967 83296049207 -198890054806 367909722520 171209435980 262657451777 -249052831865 226077432429 169083775 294195889733 -196120512403 272036969461
-248268730796 -420047016158 15793470609 9157733345 218364205063 -237031403813 468621476606 -2102355871 62299940079 347801739349 -270078157981 -169919490958
6700865680 -262634539893 -171984518254 86263634694 -1108034420 -59316508427 -75402655809 172741438676 -451535922913 -77529139537 -227073380304 375771769267
-480013940623 341132097239 -215588733107 -244393452433 704338039 -483717889274 -216244345035 285778665398 17645174 -467177163641 -171216533009 201207300803
288588217339 -386155038649 -208414271309 -297789829370 889629410 249945427615 405156403153 640144076956 1178664534 -218359538294 458333180348 85957899212
-150683532691 483995422343 -321243722865 -55352046972 6805737601 -511235753844 -79317922504 281385911039 153566501 365258939647 633365474594 505567649703
-214412069828 -80379905480 -104358589261 -32240753357 77045123 -268270107786 -327252531657 -120071100407 9736253721 201342556840 137596033128 -294472007405
188755189609 -223217089435 -245654351936 37467173567 719215590282 -164252910059 -1804918994 -27400287367 7594635792 68534515725 26666556436 100426483822
-203887274439 -42660432168 -365483262705 -52851196367 8943038368 -20657881186 207552638481 269082410527 272757961202 -73024897523 304813818491 -362010544191
55392534654 82563010463 -265105483612 -15882406819 243445642642 -642915226242 193023151999 -364981645587 189918300 -265585832999 76257406980 18800241288
389664505183 -234355680122 -177177089624 -158096426430 971389794 -65388921112 -483409177279 -49759251 71540 5583446 105763848606 170288767239 -472305642798
-21125413261 -150013989452 -680953870021 420186496693 9579815459 -87533525406 21371535322 -1434860423 1165058123876 331634450383 6244504139 24277255778
341748612359 265518577571 466828414 -282654627214 389963681292 209116808152 118650232587 -302390098251
```